

Logic, Sets and Numbers

Logic

**Def. 1.** A *mathematical statement* is an assertion that is strictly true or false. Its negation is  $\neg A := A$  is false. A statement depending on variables is a *predicate*.

**Ex. 2.**

$$P(x) := x > 2$$

**Def. 3.** Quantifiers are short hand notation for making statements about predicates.

forall :  $\forall x \in \mathbb{R} : P(x)$   
exists :  $\exists x \in \mathbb{R} : P(x)$   
exists unique :  $\exists! x \in \mathbb{R} : P(x)$   
not exists :  $\nexists x \in \mathbb{R} : P(x)$

**Imp. 4.** The oderder of quantifiers matters.  $\forall x \exists y$  is not equivalent to  $\exists y \forall x$ .

**Def. 5.** Basic logical connectives are defined as:

$A \wedge B := A$  and  $B$  are true  
 $A \vee B := A$  or  $B$  is true  
 $A \implies B := A$  implies  $B$  is true  
 $A \iff B := A \implies B \wedge B \implies A$   
Observe that  $\neg A \vee B \iff A \implies B$ .

**Thm. 6.** The following equivalences hold:

$\neg(\forall x P(x)) \iff \exists x \neg P(x)$   
 $\neg(\exists x P(x)) \iff \forall x \neg P(x)$

aswell as

$\forall x (P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$   
 $\exists x (P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$

Sets

**Def. 7.** A *set* is a collection of elements. The empty set  $\emptyset$  contains no elements.  $x \in A$  ( $x \notin A$ ) means  $x$  is (not) in  $A$ .  $A$  is a subset of  $B$  denoted by  $A \subseteq B$  iff.

$\forall x (x \in A \implies x \in B)$

$A$  is a proper subset of  $B$  denoted by  $A \subset B$  iff.  
 $A \subseteq B \wedge A \neq B$

**Def. 8.** The *union* of  $A$  and  $B$  is defined as  $A \cup B := \{x \mid x \in A \vee x \in B\}$ .

**Def. 9.** The *intersection* of  $A$  and  $B$  is defined as  $A \cap B := \{x \mid x \in A \wedge x \in B\}$ .

**Def. 10.** The *difference* of  $A$  and  $B$  is defined as  $A \setminus B := \{x \mid x \in A \wedge x \notin B\}$ .

**Def. 11.** The *Cartesian product* of  $A$  and  $B$  is defined as  $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$ .

Numbers

The basic and often used sets of numbers are:

- $\mathbb{N}$ : The set of natural numbers 1, 2, 3, 4, ...
- $\mathbb{Z}$ : The set of integers ... -2, -1, 0, 1, 2, ...
- $\mathbb{Q}$ : The set of rational numbers  $\frac{p}{q}$  where  $p, q \in \mathbb{Z}$
- $\mathbb{R}$ : The set of real numbers

**Rem. 12.** Some extensions of the real numbers are

$\mathbb{R}_0^+ / \mathbb{R}^+ := [0, \infty) / (0, \infty)$   
 $\mathbb{R}_0^- / \mathbb{R}^- := (-\infty, 0] / (-\infty, 0)$

**Def. 13.** Continuous subsets are called *intervalls*. An intervall is called

**Def. 14.** *open* if it does not contain its end-points

$I = (a, b) := \{x \mid a < x < b\}$

**Def. 15.** *closed* if it contains its endpoints

$I = [a, b] := \{x \mid a \leq x \leq b\}$

**Def. 16.** *half open* if it contains one endpoint and not the other

$I = [a, b) := \{x \mid a \leq x < b\}$

**Def. 17.** An intervall is *bounded* if it is bounded from above and below hence  $a$  and  $b$  are finite. A bounded and closed intervall is *compact*.

**Def. 18.** An *upper (lower) bound* of a subset  $X \subset \mathbb{R}$  is an element  $y \in \mathbb{R}$  such that:

$\forall x \in X : x \leq (\geq) y$

**Def. 19.** The *maximum (minimum)* of a

subset  $X \subset \mathbb{R}$  is the element  $x_0 \in X$  such that:

$\forall x \in X : x \leq (\geq) x_0$

**Def. 20.** Least upper bound and greatest lower bound are also called *Supremum* and *Infimum*.

The *Supremum*  $\sup(X)$  of a subset  $X \subset \mathbb{R}$  is the least upper bound of  $X$ . and can be characterized by:

$\forall x \in X : x \leq \sup(X) \wedge$

$\forall \epsilon > 0 : \exists x \in X : x > \sup(X) - \epsilon$

The *Infimum*  $\inf(X)$  of a subset  $X \subset \mathbb{R}$  is the greatest lower bound of  $X$ . and can be characterized by:

$\forall x \in X : x \geq \inf(X) \wedge$

$\forall \epsilon > 0 : \exists x \in X : x < \inf(X) + \epsilon$

**Thm. 21.** The *maximum and minimum* are *unique* aslong as they exist.

Every *non empty, bounded from above (below)* subset of  $\mathbb{R}$  has a *maximum (minimum)*.

**Def. 22.** A set  $X$  is said to be (order)complete if for every non empty subsets  $A, B \subseteq X$

$\forall a \in A \forall b \in B : a \leq b$

then there there exists an element  $c \in X$  such that:

$\forall a \in A : a \leq c \wedge \forall b \in B : c \leq b$

**Thm. 23.** The *Archimedian Principle* says that

$\forall x \in \mathbb{R}, \forall y > 0 \exists n \in \mathbb{N} : ny > x$   
 $\forall \epsilon > 0 \exists n \in \mathbb{N} : \frac{1}{n} < \epsilon$

**Thm. 24.** The *Young Inequality* states that for all  $\epsilon > 0$  and all  $x, y \in \mathbb{R}$ :

$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$

**Thm. 25.**  $\mathbb{R}$  has order axioms: *reflexivity, transitivity, antisymmetry, totality, making it an ordered and order-complete field.*

**Def. 26.** For  $x, y \in \mathbb{R}^n$ , the standard scalar

product is defined as:

$x \cdot y = \langle x, y \rangle := \sum_{i=1}^n x_i y_i$

**Def. 27.** The *Euclidean norm* and *distance* are defined as:

$||x|| := \sqrt{\sum_{i=1}^n x_i^2}$   
 $d(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

**Thm. 28.** The *triangle inequality* holds for all  $x, y, z \in \mathbb{R}^n$ :

$||x - z|| \leq ||x - y|| + ||y - z||$

**Def. 29.** The *complex numbers*  $\mathbb{C}$  are numbers of the form:

$z = a + ib$  where  $a, b \in \mathbb{R}$

$a$  is the real part  $\text{Re}(z)$ , and  $b$  is the imaginary part  $\text{Im}(z)$ . The imaginary unit  $i$  has the property  $i^2 = -1$ .

**Def. 30.** Arithmetic of complex numbers  $z_1 = x_1 + iy_1$  and  $z_2 = x_2 + iy_2$ :

$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$   
 $z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)$

**Def. 31.** The *complex conjugate* of  $z = x + iy$  is  $\bar{z} = x - iy$ . Division is calculated by expanding with the conjugate of the denominator:

$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$

**Rem. 32.** Some usfull properties of the conjugate:

- $z \cdot \bar{z} = |z|^2$
- $z + \bar{z} = 2\text{Re}(z)$  and  $z - \bar{z} = 2i\text{Im}(z)$
- $\overline{z + w} = \bar{z} + \bar{w}$  and  $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

**Def. 33.** Geometrically, complex numbers can be represented in the Gaussian plane. The **absolute value** of  $z = a + ib$  is its **distance from the origin**:

$$|z| = \sqrt{a^2 + b^2}$$

The distance between  $z_1$  and  $z_2$  is  $d = |z_1 - z_2|$ . The triangle inequality holds in  $\mathbb{C}$ :  $|z + w| \leq |z| + |w|$ .

**Thm. 34.** The Euler's formula states that for  $t \in \mathbb{R}$ :

$$e^{it} = \cos(t) + i \sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Eulers formula leads us to a new representation of complex numbers: the polar-coordinates.

**Def. 35.** For  $z \in \mathbb{C}$ , the polar-coordinates are defined as:

$$z = r(\cos(t) + i \sin(t)) = re^{it}$$

where  $r = |z|$  is the absolute value and  $t = \arg(z)$  is the argument.

Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1 + t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1 - t_2)}$$

So is exponentiation or taking roots:

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i \frac{t + 2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

**Def. 36.** The  $n$ -th roots of unity are the solutions to  $z^n = 1$ , which are given by:

$$z_k = e^{i \frac{2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

**Thm. 37.** The sum of the roots of unity is always zero.

$$\sum_{k=0}^{n-1} e^{i \frac{2\pi k}{n}} = 0$$

**Thm. 38.** The fundamental theorem of algebra states that every polynomial of degree  $n \geq 1$  has exactly  $n$  roots in  $\mathbb{C}$  (counting multiplicity). These roots appear in konplex-conjugate pairs.

## Sequences and Series

### Sequences

**Def. 39.** A **sequence** (Folge)  $(a_n)_{n \in \mathbb{N}_0}$  is a list of numbers, mathematically a mapping

$$f: \mathbb{N}_0 \rightarrow X$$

It can be given explicitly ( $a_n = f(n)$ ) or recursively ( $a_n = g(a_{n-1}, a_{n-2}, \dots)$ ).

**Def. 40.** A sequence  $(a_n)$  is called **convergent** with limit  $L \in \mathbb{R}$  if:

$$\forall \epsilon > 0 \exists N > 0 \text{ s.t. } \forall n > N : |a_n - L| < \epsilon$$

Written as  $\lim_{n \rightarrow \infty} a_n = L$ .

If no such  $L$  exists, the sequence is **divergent**. Sequences that diverge to  $\pm\infty$  are a special case of divergent sequences.

**Thm. 41.** The limit of a sequence is unique aslong as it exists.

**Thm. 42.** A subsequence  $(a_{n_k})_{k \in \mathbb{N}_0}$  of a convergent sequence converges to the same limit  $L$ .

**Def. 43.** An **accumulation point** (Häufungspunkt)  $A \in \mathbb{R}$  of a sequence fulfills:

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \epsilon$$

$A$  is an accumulation point iff there exists a convergent subsequence with limit  $A$ .

**Thm. 44.** Limit arithmetic: For  $\lim a_n = K$  and  $\lim b_n = L$ :

- $\lim(a_n \pm b_n) = K \pm L$
- $\lim(a_n \cdot b_n) = K \cdot L$
- $\lim(a_n/b_n) = K/L$  (if  $L \neq 0$  and  $b_n \neq 0$ )

**Thm. 45. Sandwich-Theorem:** If  $a_n \leq b_n \leq c_n$  for  $n \geq N_0$ , and  $\lim a_n = \lim c_n = L$ , then  $\lim b_n = L$ .

**Rem. 46.** Important limits:

- $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$
- $\lim_{n \rightarrow \infty} n^{1/n} = 1$
- $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$

**Def. 47.** A sequence is **monotonically increasing** (decreasing) if

$$a_n \leq (\geq) a_m \quad \forall m > n$$

and **upper** (lower) **bounded** if there exists  $M$  such that

$$|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$$

**Thm. 48. Bolzano-Weierstrass Theorem:** Every bounded sequence has an accumulation point.

**Thm. 49. Monotone Convergence Theorem:** Every bounded monotonic sequence converges to its supremum (infimum).

**Def. 50.** The limit of the supremum is called **limes superior** and written as:

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$$

The limit of the infimum is called **limes inferior** and written as:

$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k$$

**Def. 51.** A sequence is a **Cauchy sequence** if

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \epsilon$$

**Thm. 52.** A sequence of real numbers converges iff it is a Cauchy sequence.

### Series

**Def. 53.** A **series** (Reihe) is an infinite sum  $\sum_{i=0}^{\infty} a_i$ . It converges iff its sequence of partial sums converges.

**Thm. 54. Necessary condition for convergence**

$$\lim_{n \rightarrow \infty} a_n = 0$$

**Rem. 55. Geometric series:**  $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$  iff  $|x| < 1$ . **Harmonic series:**  $\sum_{n=1}^{\infty} \frac{1}{n^s}$  converges for  $s > 1$ .

**Thm. 56.** For series with non-negative terms  $a_n \geq 0$ , the sequence of partial sums is monotonically increasing. It converges iff it is bounded.

**Thm. 57. Comparison test** (Vergleichskriterium): If  $c_n \leq b_n \leq a_n$  for  $n \geq N$ :

If  $\sum a_n$  converges, then  $\sum b_n$  converges (**Majorantenkriterium**).

If  $\sum c_n$  diverges, then  $\sum b_n$  diverges (**Minorantenkriterium**).

**Def. 58.** A series is **absolutely convergent** if  $\sum |a_n|$  converges.

A **conditionally convergent** series converges, but not absolutely.

**Thm. 59. Riemann's rearrangement theorem:** Let  $\sum a_n$  be a conditionally convergent series. Then for any  $x \in \mathbb{R}$ , there exists a rearrangement of the series that converges to  $x$ .

**Thm. 60. Leibniz Criterion:** For an alternating series  $\sum (-1)^n a_n$  with a monotonically decreasing null sequence  $(a_n)$ , the series converges.

**Thm. 61. Cauchy Criterion:** The series  $\sum a_n$  converges iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : \left| \sum_{k=n}^m a_k \right| < \epsilon$$

**Thm. 62. Root Criterion** (Wurzelkriterium):

$$\rho = \limsup \sqrt[n]{|a_n|}$$

If  $\rho < 1$ , abs conv; if  $\rho > 1$ , div.

**Thm. 63. Ratio Criterion** (Quotientenkriterium):

$$\rho = \lim \left| \frac{a_{n+1}}{a_n} \right|$$

If  $\rho < 1$ , abs conv; if  $\rho > 1$ , div.

**Def. 64.** A **power series** (Potenzreihe) centered at  $a$  has the form  $\sum_{k=0}^{\infty} c_k (x - a)^k$ .

**Thm. 65.** Every power series has a **convergence radius**  $R$ , with the **interval of convergence**  $(a - R, a + R)$ .  $R$  is computed by the formula  $R = \frac{1}{\limsup |c_n|^{1/n}}$ .

## Functions

**Def. 66.** A **function**  $f : D \rightarrow C$  is a map from a domain  $D$  to a codomain  $C$  s.t. for each  $x \in D$  there is exactly one  $y \in C$  s.t.  $f(x) = y$ .

**Rem. 67.** The input is called the **independent variable** (argument) while the output is called the **dependent variable**.

**Def. 68.** A function  $f$  is called **injective** iff

$$\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$$

**Def. 69.** A function  $f$  is called **surjective** iff

$$\forall y \in B \exists x \in A : f(x) = y$$

**Def. 70.** A function  $f$  is called **bijective** iff  $f$  is injective and surjective.

**Def. 71.** The **restriction** of  $f$  under  $D' \subset \mathbb{D}$  is defined as

$$f|_{D'} : D' \rightarrow C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$$

**Def. 72.** Let  $f : D \rightarrow C$  and  $g : C \rightarrow B$  be functions then

$$g \circ f : D \rightarrow B \quad \text{with}$$

$$g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the **composition** of  $f$  and  $g$ .

**Def. 73.** The **identity function**  $id_D : D \rightarrow D$  is defined as  $id_D(x) = x \quad \forall x \in D$ .

**Def. 74.** If a function  $f : D \rightarrow C$  is bijective then there exists a unique function  $f^{-1} : C \rightarrow D$  such that

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C \quad \text{and}$$

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

$f^{-1}$  is called the **inverse function** of  $f$ .

**Def. 75.** A function  $f$  is called **bounded** if  $\exists M \in \mathbb{R}$  such that  $|f(x)| \leq (\geq) M \quad \forall x \in D$ .

**Def. 76.** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called (strictly) **monotonically increasing (decreasing)** iff

$$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < (>) f(x_2)$$

**Thm. 77.** Every strictly monotonic function is injective.

**Def. 78.** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called

- **even** iff  $\forall x \in \mathbb{R} : f(-x) = f(x)$
- **odd** iff  $\forall x \in \mathbb{R} : f(-x) = -f(x)$

**Def. 79.** The **graph** of a function  $f : D \rightarrow C$  is the set of all ordered pairs  $(x, f(x))$  for  $x \in D$ .

**Def. 80.** A graph is **konvex** if it lies below the line segment connecting any two points on the graph. Similarly it is called **konkav** if it lies above the line segment connecting any two points on the graph.

## Limits in the context of functions

**Def. 81.** Let  $f : \mathbb{D} \rightarrow \mathbb{R}$  be a function satisfying  $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$  for all  $\delta > 0$  and  $x_0 \in \mathbb{R}$ . We say that  $\lim_{x \rightarrow x_0} f(x) = L$  iff

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t.}$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$

The limit  $L$  is unique.

**Def. 82.** A function  $f : \mathbb{D} \rightarrow \mathbb{R}$  has in  $x_0$  a **right-sided limit**  $\lim_{x \searrow x_0} f(x) = \lim_{x \rightarrow x_0^+} f(x) = L$  iff

$$\forall \varepsilon > 0 \exists \delta > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

**left-sided limit**  $\lim_{x \nearrow x_0} f(x) = L$  iff

$$\forall \varepsilon > 0 \exists \delta > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon)$$

**Def. 83.** A function  $f : \mathbb{D} \rightarrow \mathbb{R}$  has a **limit for  $x$  approaching infinity**  $\lim_{x \rightarrow \infty} f(x) = L$  iff

$$\forall \varepsilon > 0 \exists M > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(x > M \implies |f(x) - L| < \varepsilon)$$

**limit for  $x$  approaching negative infinity**  $\lim_{x \rightarrow -\infty} f(x) = L$  iff

$$\forall \varepsilon > 0 \exists M > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(x < -M \implies |f(x) - L| < \varepsilon)$$

**Def. 84.** A function  $f : \mathbb{D} \rightarrow \mathbb{R}$  has in  $x_0$  a **improper limit approaching infinity**  $\lim_{x \rightarrow x_0} f(x) = \infty$  iff

$$\forall N > 0 \exists \delta > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$

**improper limit approaching negative infinity**  $\lim_{x \rightarrow x_0} f(x) = -\infty$  iff

$$\forall N > 0 \exists \delta > 0 \quad \text{s.t.}$$

$$\forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) < -N)$$

**Thm. 85.** Let  $A \in \mathbb{R}$  and assume  $\lim_{x \rightarrow x_0} f(x) = K (\neq \pm\infty)$  and  $\lim_{x \rightarrow x_0} g(x) = L (\neq \pm\infty)$ .

- $\lim_{x \rightarrow x_0} (f(x) + g(x)) = K + L$
- $\lim_{x \rightarrow x_0} (f(x) - g(x)) = K - L$
- $\lim_{x \rightarrow x_0} (f(x)g(x)) = KL$
- $\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L$

$$\bullet \lim_{x \rightarrow x_0} (Af(x)) = AK$$

$$\bullet \lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L \text{ if } L \neq 0$$

## Continuity

**Def. 86.** A function  $f : D \rightarrow \mathbb{R}$  is **continous** at  $x_0$  if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } \forall x \in D$$

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon$$

$f$  is **continuous** iff for every sequence  $x_n \rightarrow x_0$ ,  $f(x_n) \rightarrow f(x_0)$ .

**Thm. 87.** If  $f, g$  are continuous, then  $f \pm g$ ,  $f \cdot g$ ,  $c \cdot f$ ,  $f/g$  (if  $g \neq 0$ ), and  $f \circ g$  are also continuous.

**Thm. 88.** If  $f : I \rightarrow J$  is continuous and bijective,  $J$  is an interval and  $f^{-1} : J \rightarrow I$  is continuous.

**Thm. 89. Intermediate value theorem:** Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. For any  $c \in [f(a), f(b)]$ , there is an  $x_0 \in [a, b]$  with  $f(x_0) = c$ .

**Def. 90.**  $f$  has a **local maximum (minimum)** at  $x_0$  if  $f(x_0) \geq (\leq) f(x) \quad \forall x \in D$ . Both are called **extrema**.

**Thm. 91.** A continuous function on a **compact interval** is bounded and attains its maximum and minimum value.

## Differential calculus

**Def. 92.** The **average rate of change** of a function  $f$  in the domain between  $a$  and  $a + h$  is defined as

$$\frac{\Delta y}{\Delta x} = \frac{f(a + h) - f(a)}{h}$$

and also called the differential quotient.

**Def. 93.** The **derivative**  $f'$  of a function  $f$  at the coordinate  $a$  is defined as

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} &= \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a} \\ &= \frac{df}{dx} \Big|_{x=a} = f'(a) \end{aligned}$$

**Def. 94.** If for a function  $f$  at the coordinate  $a$  the derivative  $\exists f'(a)$  then  $f$  is called **differentiable** at  $a$ . If  $f$  is differentiable for every  $a \in \mathbb{D}$  then  $f$  is called differentiable.

**Lem. 95.** If  $f$  is differentiable at  $x_0$  then  $f$  is continous at  $x_0$ .

**Def. 96.** If the limit

$$\lim_{h>0 \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

existst then we call it the **right-sided derivative** (or the left-sided derivative analogue)

**Def. 97.** For a function  $f$  we can iteratively define higher derivatives for  $n \in \mathbb{N}_0$

$$f^{(0)} = f, f^{(1)} = f', f^{(2)} = f'', \dots$$

$$f^{(n+1)} = (f^{(n)})'$$

if  $f^{(n)}$  exists we call it  $n$ -times differentiable.

**Def. 98.** if  $f^{(n)}$  is continous we call  $f$   $n$ -times continous differentiable. The set of all  $n$ -times continous differentiable functions on  $D$  is denoted as  $C^n(D)$

**Def. 99.** We call  $C^\infty(D) := \cap_{n=0}^\infty C^n(D)$  the set of smooth functions.

## Derivation rules

**Thm. 100.** For functions  $f(x)$  we have

$$\bullet \quad f(x) = ax^p \quad \implies \quad f'(x) = apx^{p-1}$$

$$\bullet \quad f(x) = a^x \quad \implies \quad f'(x) = \ln(a) \cdot a^x$$

$$\bullet \quad f(x) = \sin(x) \quad \implies \quad f'(x) = \cos(x)$$

$$\bullet \quad f(x) = \cos(x) \quad \implies \quad f'(x) = -\sin(x)$$

$$\bullet \quad f(x) = \tan(x) \quad \implies \quad f'(x) = \frac{1}{\cos^2(x)} = 1 + \tan^2(x)$$

$$\bullet \quad f(x) = \sinh(x) \quad \implies \quad f'(x) = \cosh(x)$$

$$\bullet \quad f(x) = \cosh(x) \quad \implies \quad f'(x) = \sinh(x)$$

$$\bullet \quad f(x) = \ln(x) \quad \implies \quad f'(x) = \frac{1}{x}$$

$$\bullet \quad f(x) = \arcsin(x) \quad \implies \quad f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\bullet \quad f(x) = \arccos(x) \quad \implies \quad f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\bullet \quad f(x) = \arctan(x) \quad \implies \quad f'(x) = \frac{1}{1+x^2}$$

**Thm. 101. Sum Rule:** For the  $n$ -th deriva-

tive of a sum:

$$(f+g)^{(n)} = f^{(n)} + g^{(n)}$$

**Thm. 102. General Leibniz Rule (Product):** The  $n$ -th derivative of a product is given by:

$$(f \cdot g)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}$$

**Thm. 103. Chain Rule:** For the composition of functions:

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

**Thm. 104. Quotient Rule:** For  $g(x) \neq 0$ :

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

**Thm. 105. Inverse Functions:** If  $f$  is differentiable and injective:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

## Bernoulli-L'Hopital

**Thm. 106.** Let  $I \subset \mathbb{R}$  be a an intervall,  $f : I \rightarrow \mathbb{R}$ , and let  $x_0$  be a local Extrema of  $f$ . Then one of the following holds:

- $x_0$  is an endpoint of  $I$
- $f$  is not differentiable at  $x_0$
- $f$  is differentiable at  $x_0$  and  $f'(x_0) = 0$

**Thm. 107. Rolle's Theorem:** Let  $f : [a, b] \rightarrow \mathbb{R}$  be differentiable on  $(a, b)$  and continous on  $[a, b]$ , and assume that  $f(a) = f(b)$ , then there exists a  $\xi \in (a, b)$  such that  $f'(\xi) = 0$

**Thm. 108. Mean Value Theorem:** Let  $f : [a, b] \rightarrow \mathbb{R}$  be differentiable on  $(a, b)$  and continous on  $[a, b]$ , then there exists a  $\xi \in (a, b)$  such that  $f'(\xi) = \frac{f(b)-f(a)}{b-a}$

**Thm. 109. L'Hopital's Rule:** Let  $f, g : I \rightarrow \mathbb{R}$  differentiable functions on  $I \subset \mathbb{R}$  and  $a \in \bar{I}$ . If  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \in \{0, \infty\}$  ( $0/0$  or  $\inf/\inf$ ) and  $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$  exists then  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$

## Taylor Approximation

**Def. 110.** A function  $f$  is called **concave** / **convex** on  $I$  if for all  $a, b \in I$  and  $t \in (0, 1)$

$$f((1-t)a + tb) \geq (\leq) (1-t)f(a) + tf(b)$$

**Thm. 111.**  $f$  is convex/concave iff  $f'$  is monotonically increasing/decreasing on  $I$  i.e. if  $f'' \geq (\leq) 0$

**Thm. 112. Taylor's Theorem:** Let  $f : (a, b) \rightarrow \mathbb{R}$  be  $n+1$ -times differentiable, then for all  $x \in (a, b)$  then  $f(x) = P_n(x) + R_n(x)$  where  $P_n(x)$  is the Taylor polynomial of degree  $n$  at  $a$

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

and  $R_n(x)$  is the remainder term

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{n+1!} (x-a)^{n+1}$$

for some  $\xi \in (a, b)$